2025 CSC493 – Capstone Weekly Reports[1]

<Download this template and name it Week_<WEEK_NUMBER>, where "<REPORT_NUMBER>" is replaced by the current week number (i.e., "1" for the first weekly report and so forth).

Week	Week 5
Project Name	AI Music Mood Classifier
Report Date	Sep 23, 2025 ← click to select date

Part 1: Weekly Progress Report

- **Accomplishments**: **What did you complete this week?** Be specific and measurable.
- Examples: *Implemented login authentication using OAuth2; Fixed bug in file upload function (issues #14); Completed draft of UI wireframes; Identified functional and non-functional requirements; etc. (up to 3 points)*

This week, I added a basic skeleton for my GUI file in order to have a simple visual that I can connect my working model to later, in order for users to upload music files and have the AI model classify the mood of the song. I also added an output for a features.csv file so I don't have to constantly extract the features from the DEAM dataset anytime I want to train my model on the data from it. Lastly, I added k-fold CV accuracy scoring to my logistic regression baseline model in order to determine how accurate the logistic regression model is, in order to find any issues simply and easily, and to also compare it to other models and see which model works best.

- **Challenges**: **What problems did you encounter?** Mention blockers, unresolved issues, or risks.
- Examples: *Struggled with connecting database migrations to production server; API documentation was incomplete – need clarification; etc. (up to 3 points)*

I did not run into any real challenges this week as I found some great resources on how to build a logistic regression model using the sklearn library. The only slight issue I ran into was a very simple typo mistake where I typed in the wrong variable in main.py for training the valence and arousal parts of the model; I typed in df_labels rather than dataset, which it should have been.

- **Next Steps**: **What are your goals for next week?** Keep them realistic and linked to project milestones.
- Examples: Set up automated testing with GitHub Actions; Final project proposal *(up to 2 points)*

My goals for next week are to start implementing the random forest AI model for comparing to the logistic regression model, and train both models on all songs of the DEAM dataset, and compare the two accuracy results from k-fold cross-validation score.

- **Discussion Points**: **What would you like to discuss in class, or with your professor/TA?** This could include technical issues, project direction, or collaboration concerns.
- Examples: *Should we prioritize front-end features before database scaling? Need guidance on writing integration tests; etc. (up to 2 points)*

[1] Detailed Weekly Report requirements can be found here: 2024 URCPP Capstone - Using Agile and Reporting Out

Should I add more than just accuracy, like F1-score, ROC AUC, or confusion matrices, to better capture model performance? Should I keep adding more low-level audio features (like spectral flatness, tonnetz) or focus on optimizing models with the features I already have?

Part 2: Time Reporting

Make sure that as you fill out the first prompt, you include in enough detail in the summary. For example, "debugging" is vague, but "debugged function X to make sure that when user does action Y, it is called and returns the value Z" is better.

- **Time Spent**: Briefly explain how much time, *outside of class*, spent on your project. If you worked on multiple components, each should get a detailed summary. Make sure to add up all the hours and minutes correctly. *Add as many rows as you need to the table below. Please do not include hours in class as part of this section. (up to 4 points)*

START	FINISH	HOURS	DETAILED SUMMARY
MM/DD HH:MM	MM/DD HH:MM		
9/16 08:00	9/16 12:00	4	I added an addition to main.py that outputs the extracted features from the DEAM dataset to a CSV file titled features.csv. I also started adding k-fold cross-validation for scoring the accuracy of my logistic regression model
9/17 19:00	9/17 23:00	3	Watched some tutorials on sklearn and logistic regression models to better understand how to implement it and reduce redundant code.
9/18 10:00	9/18 14:00	4	Fully added the logistic regression model to baseline_classifier and added a simple skeleton of GUI.py for a rudimentary GUI that has an upload song file button, predict mood button (that will later be connected to the final AI model), and an output text box that displays the predicted mood and other information pertaining to the prediction.
9/19 15:00	9/19 19:00	4	Watched more tutorials on logistic regression and accuracy scoring in order to understand what accuracy scores are acceptable and what can be changed with the model to improve the accuracy.
	WEEKLY TOTAL	15	

- **Total (Cumulative) Project Time Spent**: After the number of hours and minutes, make sure to briefly explain whether you are on track and if not, what you may need to do in order to achieve what you set out to accomplish. *(up to 2 points)*

So far, I have spent 66 hours total on this project. According to the timeline that I built with help from ChatGPT, I am currently around one to one and a half weeks ahead of schedule at the moment. The k-fold cross-validation for scoring model accuracy and even the model implementation were meant to start this week and be completed by the end of next week.

Rubric:

The following rubric will be used, but they might change as needed.

Accomplishments (3 points)

1 point for a general description of progress, 2 points for specifics on progress, 3 points for specifics AND referring to previous targets and explaining how current accomplishments build on previous ones.

Challenges (3 points)

1 point for mentioning there are roadblocks, 2 points for specifics, 3 points for specifics AND what was done already to try to overcome them.

Desired discussion points (2 points)

1 point for at least one relevant discussion point as a general question, 2 points for relevant discussion points with specifics

Future Goals (2 points)

1 point for concrete future targets (i.e. "working more on the project" is a zero, but "working on getting component X to interface with component Y" suffices), 2 points for tying in the targets with what was hopefully discussed in the meeting.

Time Spent (4 points)

1 point for including general statements of how much time was spent ("4 hours on coding"), 2 points for splitting time into specific parts ("1.5 hours on research on component X, 1 hour coding, 2.5 hours debugging"), 3 points for specific parts and details on the pieces ("1.5 hours researching Turtle interface for drawing concentric circles given inputs from the user, 1 hour coding function X that used that interface, 2.5 hours testing function X by giving it multiple values and fixing errors for values A, B, C, and D"). 1 Point for totalling the hours correctly.

What happens if your time on a task is interrupted and you don't have a concrete (or discrete) end time? In this case put the start time in, and the word "interrupted" for the end time and include the task total time. Rounding to the nearest 15 minutes is acceptable. *(This makes adding up times easier, especially when you use decimal hours, i.e. 3.75 rather than 3 hours 45 minutes.)*

Total (Cumulative) Project Time (2 points)

1 point for summing the values correctly, 2 points for the total time AND reflection on progress (you are confident to fit the target and if not, what course corrections you anticipate needing to make)

Resources

Here's a link to this Weekly Report Template – Make a copy and use it:

2025 CSC493 Weekly Report v2 TEMPLATE